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Carbonylation

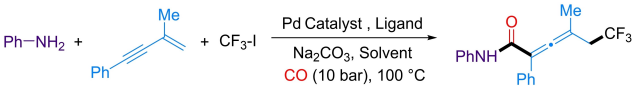
Palladium-Catalyzed Carbonylative Multicomponent Fluoroalkylation of 1,3-Enynes: Concise Construction of Diverse Cyclic Compounds

*Chang-Sheng Kuai, Bing-Hong Teng, and Xiao-Feng Wu**

Abstract: Multicomponent reactions, particularly those entailing four or more reagents, have presented a longstanding challenge due to the inherent complexities associated with balancing reactivity, selectivity, and compatibility. In this study, we describe a palladium-catalyzed multi-component fluoroalkylative carbonylation of 1,3-enynes. A series of products featuring three active functional groups-allene, fluoroalkyl, and carboxyl, were efficiently and selectively integrated in a single chemical operation. Furthermore, more intricate fluoroalkyl-substituted pyrimidinones can be constructed by simply altering the 1,3-bisnucleophilic reagent. This approach also provides a valuable strategy for the late-stage modification of naturally occurring molecules and concise construction of diverse cyclic compounds.

Allenes, serving as reactive intermediates, participate in a diverse array of organic transformations and are ubiquitously present in natural products, pharmaceuticals, and functional materials (Scheme 1A).[1] Simultaneously, the versatility of fluorine and fluoroalkyl moieties has been duly acknowledged in the realm of functional groups. The incorporation of these fluorine-laden motifs into the organic framework invariably leads to the modification of the physiological properties of the parent molecules, culminate in heightened lipophilicity, increase bioavailability, and improve metabolic stability.[2] Moreover, derivatives bearing carboxylic acid functionality are prevalent in both pharmaceutical compounds and functional materials.[3] Considering the pivotal roles that these three moieties play in the domains of organic synthesis and medicinal chemistry, their amalgamation within a singular molecular entity holds the promise of substantial and transformative advantages for

Table 1: optimization of reaction conditions.[a]



1a 2a 3a 4a

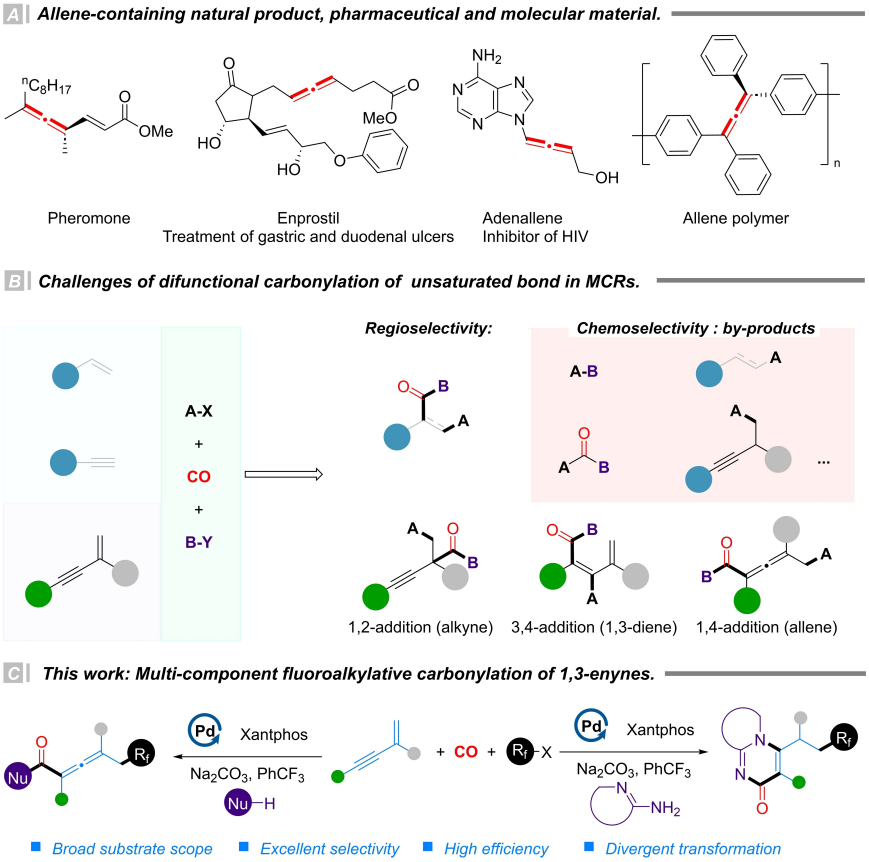
Entry	Catalyst	Ligand	Solvent	Yield **4a** [%]
1	$PdCl_2$	$BuPAd_2$	CH_3CN	NR
2	$PdCl_2$	PPh_3	CH_3CN	NR
3	$PdCl_2$	DPPE	CH_3CN	NR
4	$PdCl_2$	DPPP	CH_3CN	NR
5	$PdCl_2$	DPPF	CH_3CN	2
6	$PdCl_2$	(S,S)-Ph-BPE	CH_3CN	NR
7	$PdCl_2$	(R)-DTBM-SEGPHOS	CH_3CN	NR
8	$PdCl_2$	(R)-BINAP	CH_3CN	NR
9	$PdCl_2$	DPEphos	CH_3CN	15
10	$PdCl_2$	Xantphos	CH_3CN	57
11	$PdCl_2$	Xantphos	DCE	58
12	$PdCl_2$	Xantphos	THF	55
13	$PdCl_2$	Xantphos	DMF	31
14	$PdCl_2$	Xantphos	NMP	30
15	$PdCl_2$	Xantphos	$PhCF_3$	70
16	$Pd(OAc)_2$	Xantphos	$PhCF_3$	53
17	$Pd(PPh_3)_4$	Xantphos	$PhCF_3$	60
18	$Pd(acac)_2$	Xantphos	$PhCF_3$	55
19	$Pd(COD)Cl_2$	Xantphos	$PhCF_3$	84(82)[b]
20	–	Xantphos	$PhCF_3$	NR

[a] Reaction conditions: **1a** (0.15 mmol), **2a** (0.1 mmol), **3a** (0.2 mmol), carbon monoxide (10 bar), catalyst (5 mol%), ligand (5–10 mol%), Na_2CO_3 (0.2 mmol), solvent (1.0 mL), 100 °C, 20 h. Yield was determined by GC using ^{n}Hexadecane as the internal standard. [b] Isolated yield of **4a** provided in parentheses. $BuPAd_2$: di(1-adamantyl)-*n*-butylphosphine. DPPE: 1,2-bis(diphenylphosphino)ethane. DPPP: 1,3-bis(diphenylphosphino)propane. DPPF: 1,1′-ferrocenediyl-bis(diphenylphosphine). (S,S)-Ph-BPE: (+)-1,2-bis((2*S*,5*S*)-2,5-diphenylphospholano)ethane. (R)-DTBM-SEGPHOS: (R)-(−)-5,5′-bis-[di(3,5-di-*tert*-butyl-4-methoxyphenyl)phosphino]-4,4′-bi-1,3-benzodioxole. (R)-BINAP: (R)-(+)-2,2′-bis(diphenylphosphino)-1,1′-binaphthalene. DPEphos: bis[(2-diphenylphosphino)phenyl] ether. Xantphos: 4,5-bis(diphenylphosphino)-9,9-dimethylxanthene.

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Scheme 1. Challenges for palladium-catalyzed multicomponent fluoroalkylative carbonylation of 1,3-enynes.

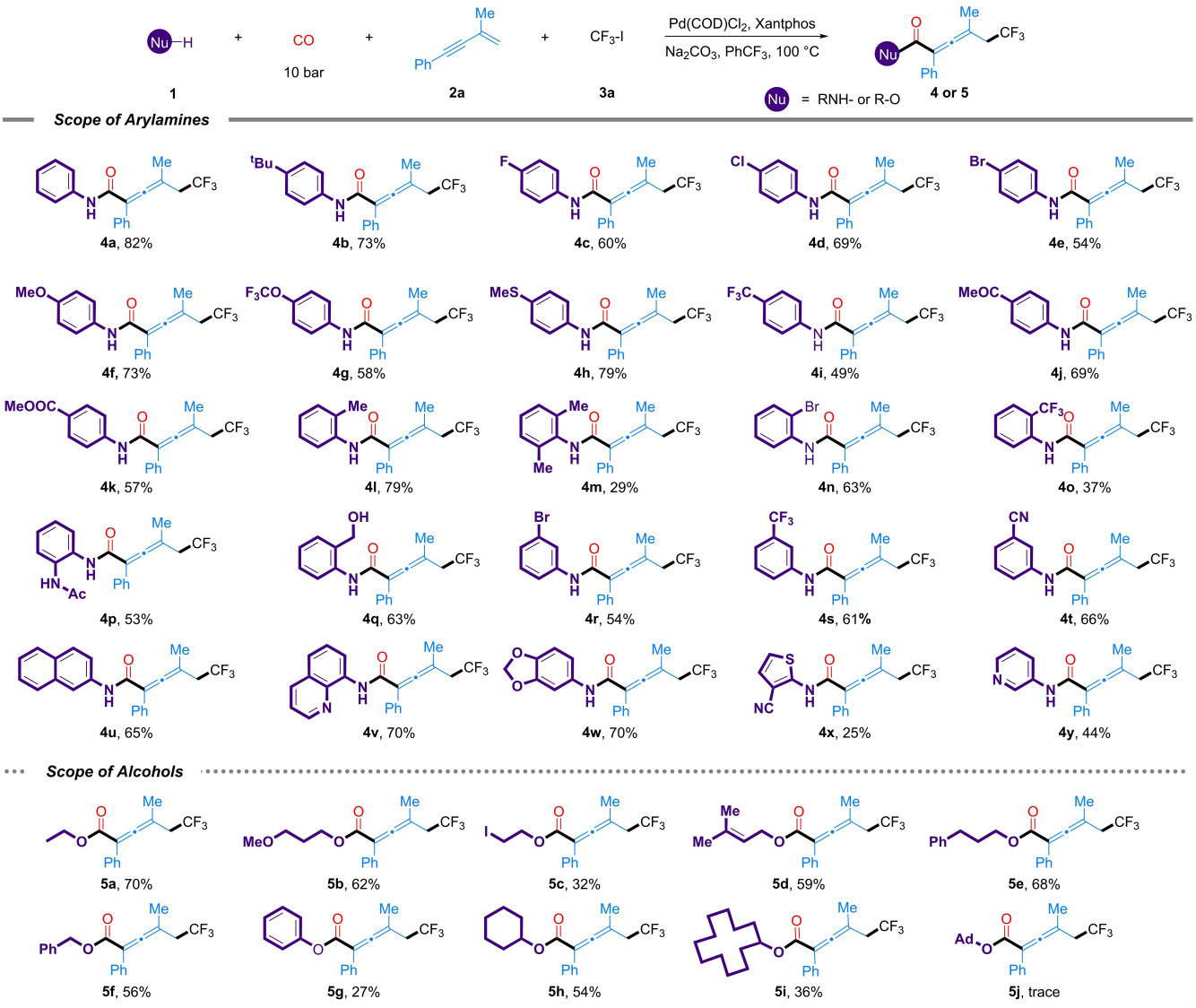
endeavors such as drug discovery, pharmaceutical synthesis, and the development of novel materials.

Multicomponent reactions (MCRs), possessing the capability to forge numerous carbon-carbon or carbon-heteroatom bonds from simple starting materials in one synthetic operation, represent a compelling strategy in organic synthesis owing to their capacity to enhance both reaction efficiency and productivity.[4] In recent years, transition metal-catalyzed carbonylation of unsaturated bonds,[3g,5] the incorporation of carbon monoxide (a pivotal bulk C1 building block) into organic molecular frameworks through multicomponent reactions (MCRs) have emerged as a potent strategy for accessing value-added compounds, including pharmaceuticals and commoditized chemicals. Recently, innovative approaches leveraging Pd-catalyzed domino perfluoroalkylation/carbonylation of olefins[6] and alkynes[7] have been devised for the synthesis of β-perfluoroalkyl-substituted unsaturated carbonyl-containing compounds. In contrast to relatively straightforward substrates such as olefins and alkynes with relatively singular selectivity (Scheme 1B), the 1,3-enyne, characterized by the simultaneous presence of both alkene and alkyne functionalities, serves as a common and valuable building block in synthetic chemistry,[8] raising the difficulties associated with the in-sequence control of the chemo-, regioselectivity. As illustrated in Scheme 1B, a minimum of three distinct regioselective structures is juxtaposed against the singular selectivity of either alkenes or alkynes.

Moreover, diverse chemoselectivity in a multicomponent reaction can concomitantly generate undesired by-products. Transition metal-catalyzed bisfunctionalization of 1,3-enynes has undergone extensive investigation.[8c,d,9] While monofunctional carbonylation of 1,3-enynes has been reported by Alper and Wu groups,[10] strategies involving domino fluoroalkylation/carbonylation of the more challenging 1,3-alkynes have remained elusive. As the continuation of our research interests in catalytic carbonylation chemistry, and considering the widespread applicability and unique properties of the alkene, fluoroalkyl, and carboxyl groups, our interest has been piqued by the potential of utilizing a multicomponent transition metal-catalyzed domino fluoroalkylation/carbonylation reaction to achieve the one-step synthesis of compounds incorporating all three of these functional groups simultaneously. To date, a streamlined synthetic method for construction of allene compounds containing both carboxyl and fluoroalkyl functionalities has yet to be replenished. Herein, we present an efficient palladium-catalyzed, highly chemo- and regioselective multicomponent fluoroalkylative carbonylation of 1,3-enynes for the synthesis of fluoroalkyl-substituted allene carboxylic derivatives. Moreover, when employing 1,3-bisnucleophilic substrates, further cyclization yields fluoroalk-

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Scheme 2. Scope of arylamines and alcohols. Reaction conditions: **1a** (0.15 mmol), **2a** (0.1 mmol), **3a** (0.2 mmol), carbon monoxide (10 bar), $Pd(COD)Cl_2$ (5 mol%), Xantphos (5 mol%), Na_2CO_3 (0.2 mmol), $PhCF_3$ (1.0 mL), 100 °C, 20 h, isolated yield.

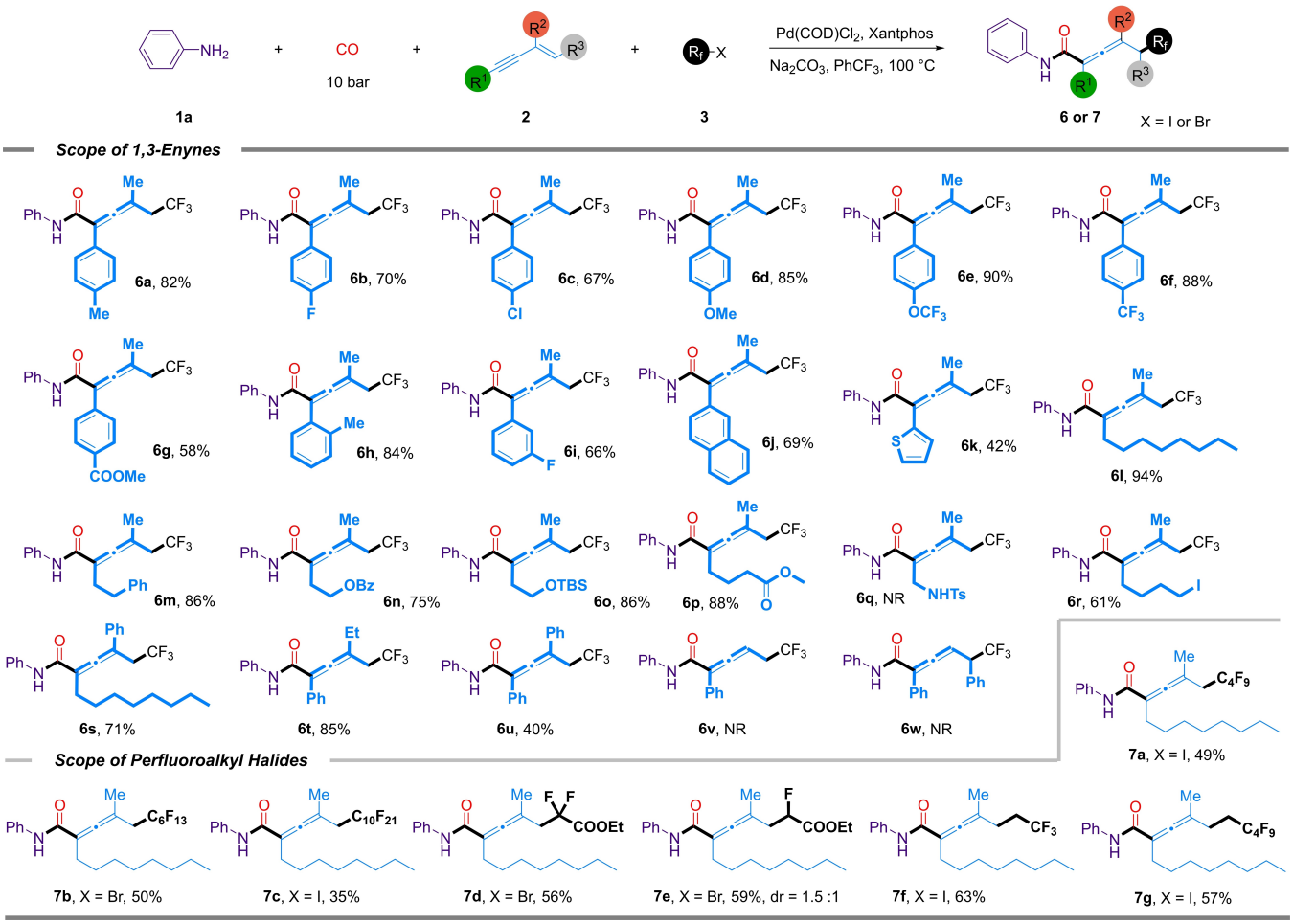
yl substituted pyrimidinone products. This approach also provides a valuable strategy for the late-stage modification of naturally occurring molecules and the concise construction of diverse cyclic compounds.

To achieve this multi-component fluoroalkylative carbonylation reaction, the investigation was initiated using aniline **1a**, 1,3-enynes **2a**, and iodotrifluoromethane **3a** as the model substrates, $PdCl_2$ as precatalyst in the presence of Na_2CO_3 in CH_3CN at 100 °C and under CO (10 bar) atmosphere (Table 1). The monophosphine ligands ($BuPAd_2$, PPh_3) proved ineffective in generating the target product (Table 1, entries 1–2). Consequently, our attention shifted towards bidentate phosphine ligands characterized by varying bite angles and even with chirality (Table 1, entries 3–10). Gratifyingly, the target product **4a** was observed in 57% yield with Xantphos as the ligand (Table 1, entry 10). Subsequently, a series of solvents were also screened (Table 1, entries 11–15). $PhCF_3$ emerged as the optimal choice, affording a superior 70% yield of **4a** compared to alternative solvents such as DCE, THF and DMF. Finally, palladium precatalyst screening revealed and $Pd(COD)Cl_2$ was found to be the best choice to give product **4a** in 84% yield (Table 1, entries 16–19). Furthermore, control experiments revealed that the catalyst was essential for the success of this reaction (Table 1, entry 20).

Having determined the optimal reaction conditions, we next explored the generality of different arylamine substrates for this multi-component fluoroalkylative carbonylation. As shown in Scheme 2, anilines bearing electron-donating groups, such as *tert*-butyl (**1b**), alkoxy (**1f**, **1g**), and even methylthiol groups (**1h**), were all tolerated under the current reaction conditions, afforded moderate to good yields (58–82%) of the desired

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Scheme 3. Scope of 1,3-enynes and perfluoroalkyl halides. Reaction conditions: **1a** (0.15 mmol), **2a** (0.1 mmol), **3a** (0.2 mmol), carbon monoxide (10 bar), $Pd(COD)Cl_2$ (5 mol%), Xantphos (5 mol%), Na_2CO_3 (0.2 mmol), $PhCF_3$ (1.0 mL), 100 °C, 20 h, isolated yield.

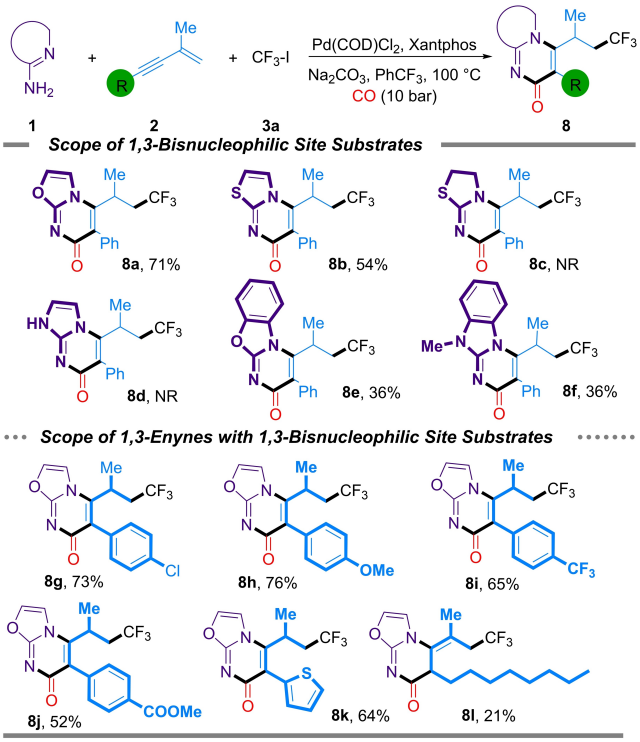
products. Additionally, various halogen atoms (F, Cl and Br) at the *para*-position of the phenyl ring were all applicable, yielding the desired products with percentages ranging from 54% to 69% (**4c**–**4e**). Anilines with electron-withdrawing groups at the *para*-position of the phenyl ring can also be applied, afforded the corresponding products in 49–57% yields (**4i**–**4k**). Due to the steric hindrance inherent in this transformation, a notable decrease in yield was observed when testing *ortho*-substituted aniline derivatives (**4l**, **4m**). Subsequently, various *ortho*- and *meta*-substituted anilines were subjected for further testing. Anilines with different functional groups, including bromo-, trifluoromethyl-, amide-, nitrile-, and even hydroxy-, were well tolerated, afforded the corresponding products **4n**–**4t** in moderate yields. Nevertheless, heterocyclic substituted amines, such as thiophene and pyridine, impede the reaction and delivers the target products in moderate yields (**4x**, **4y**). However, no desired product could be detected when *N*-methylaniline, alkyl amines or benzamide were tested as the reaction partners which most likely due to the not matched basicity.

Subsequently, with alcohols as the nucleophile, a series of fluoroalkyl-substituted allene esters were synthesized in moderate yields (**5a**–**5i**). For instance, substrates containing functional groups such as methoxy, iodide, and olefins exhibit compatibility with this process. Moreover, potentially attributed to the phenol's capacity to quench free radicals, the attainment of the target product was limited to 27% yield. In addition to the primary alcohols, secondary alcohols are also tolerated and deliver the corresponding products (**5h**, **5i**). However, tertiary alcohol, characterized by high steric hindrance, yield only trace amounts of product **5j**.

Subsequently, we turned our attention to the substrate scope of 1,3-enyne derivatives, which is shown in Scheme 3. Various aryl-substituted (R^1) 2-methyl-1,3-enynes underwent this reaction successfully under our standard conditions. Substrates featuring electron-donating groups (Me, F, Cl, OMe, OCF_3) or electron-withdrawing groups (CF_3, COOMe) on the phenyl ring were all well-tolerated, yielding the corresponding products in the range of 58% to 90% yields (**6a**–**6j**). Furthermore, the electron-deficient heterocyclic thiophene-substituted enyne was also tolerated in this transformation, yielding

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Scheme 4. Scope of 1,3-bisnucleophilic site substrates. Reaction conditions: **1a** (0.15 mmol), **2a** (0.1 mmol), **3a** (0.2 mmol), carbon monoxide (10 bar), $Pd(COD)Cl_2$ (5 mol%), Xantphos (5 mol%), Na_2CO_3 (0.2 mmol), $PhCF_3$ (1.0 mL), 100 °C, 20 h, isolated yield.

the target product **6k** in 42% yield. Next, alkyl (octyl, phenylpropyl) and diverse heteroatom-substituted alkyl groups (OBz, OTBS and OAc) on the R^1-position of 1,3-enynes were also smoothly converted to the corresponding products in 61% to 94% yields. Nevertheless, alkyl-substituted sulfonamide group (**6q**) did not exhibit reactivity in this transformation. Then, we investigated the impact of varying R^2 groups in this transformation. When R^2 was altered to phenyl or ethyl, the corresponding products **6s** and **6t** were produced in 71% and 85% yields, respectively. However, the yield of the diphenyl-substituted 1,3-enyne (**6u**) decreased to 40%. When R^2 and R^3 were hydrogen and phenyl, no reactivity was observed (**6v**, **6w**). This distinct reactivity may arise from the instability of the formed propargyl radical and the derived allenyl radical.

Finally, a variety of fluoroalkyl halides, including diverse polyfluorinated long-chain alkyl iodide or bromine substrates, bromodifluoroacetate, and ethyl 2-bromo-2-fluoroacetate, underwent smooth reactions, yielding the corresponding products **7a**–**7e** in moderate yields. Additionally, it was noteworthy to discover that perfluoroiodopropane (**7f**, **7g**) could also participate in the reaction and afforded 57% to 63% yields of the corresponding products. However, no desired product was detected when 1-fluoro-3-iodopropane was tested as the substrate.

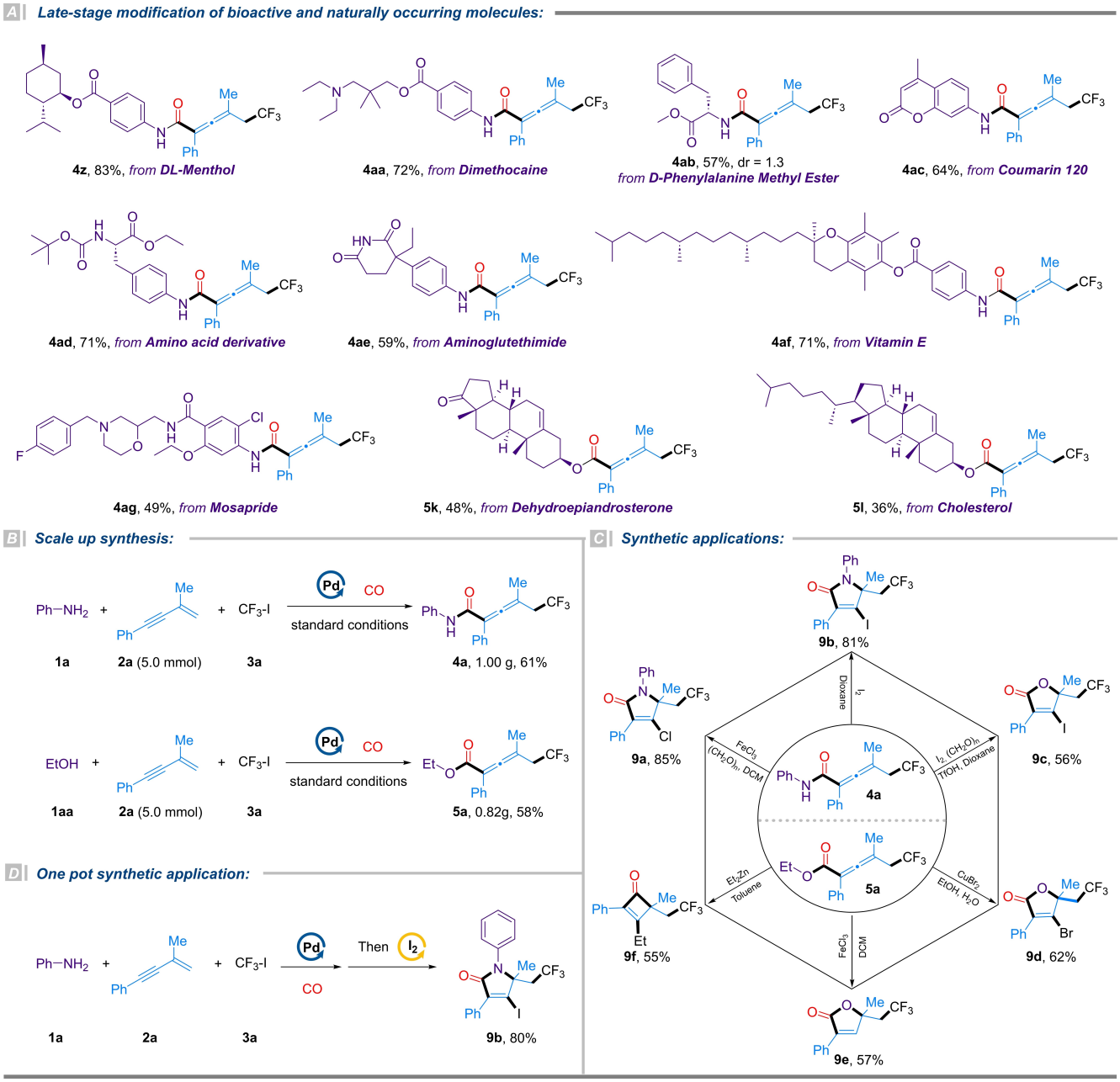
Given that allenes are a highly reactive class of substrates, we envisaged the possibility of employing bisnucleophilic substrates for subsequent reactions with allenes after the fluoroalkylative carbonylation of 1,3-enynes, aiming to obtain more complex products. Initially, we employed bisnucleophilic substrates such as guanidine, urea, thiourea and diol, but none of them exhibited reactivity. We speculate that this might be attributed to the hindrance of the initial step of fluoroalkylative carbonylation of 1,3-enynes. Subsequently, we turn to potentially bisnucleophilic heterocyclic aromatic amine substrates (Scheme 4). To our delight, when oxazol-2-amine was employed, further cyclization of the product pyrimidinone (**8a**) could be obtained in 70% yield without further optimization the reaction conditions. Other heterocyclic amines, such as thiazol-2-amine, oxazol-2-amine and *N*-methyl-benzoimidazol-2-amine, were all well compatible in the reaction (**8b**, **8e** and **8f**). However, saturated 4,5-dihydrothiazol-2-amine and 1*H*-imidazol-2-amine exhibit no reactivity (**8c**, **8d**), and we speculate that this may be attributed to their strong alkalinity, causing the reaction to stagnate. Afterwards, we shifted our focus to the compatibility of 1,3-alkyne derivatives in this conversion. Various aryl-substituted 2-methyl-1, 3-enes are involved in this reaction without any problem. Both substrates containing electron-donating groups and electron-withdrawing groups in the phenyl ring were all tolerated, and the yields of corresponding products is between 52% and 76% (**8g**–**8k**). However, when the alkyl-substituted 1, 3-alkyne was employed in the reaction, the alkene isomerization product **8l** was obtained in 21%.

To further illustrate the synthetic utility of this strategy, several experiments were conducted as depicted in Scheme 5. Initially, various natural products and bioactive molecule modified amines and alcohols were investigated to verify the practicability of this method. Gratifyingly, biologically active and naturally occurring amines, such as DL-menthol (**4z**), dimethocaine (**4aa**), amino acid derivatives (**4ab**, **4ad**), coumarin 120 (**4ac**) aminoglutethimide (**4ae**), Vitamin E (**4af**), and mosapride (**4ag**), were all successfully converted into the corresponding products in 49%–83% yields (Scheme 5A). Furthermore, bioactive alcohols such as dehydroepiandrosterone (**5k**) and cholesterol (**5l**) showed low reactivity, afforded the corresponding products in 48% and 36% yields, respectively. The successful applications of this method to these complex molecules will have significant implications for the discovery and research of drugs and materials. Next, two scale-up transformations with aniline and ethanol gave acceptable yields (**4a**, 61%; **5a**, 58%), respectively (Scheme 5B). The fluoroalkyl substituted allene carboxylic derivatives, as useful synthetic intermediates, can be converted to other versatile chemicals. Thus, concise and divergent constructions of various cyclic compounds have been demonstrated, as shown in Scheme 5C.

Treatment of **4a** with a mixture of $FeCl_3$ and paraformaldehyde or I_2 successfully affording chloro/

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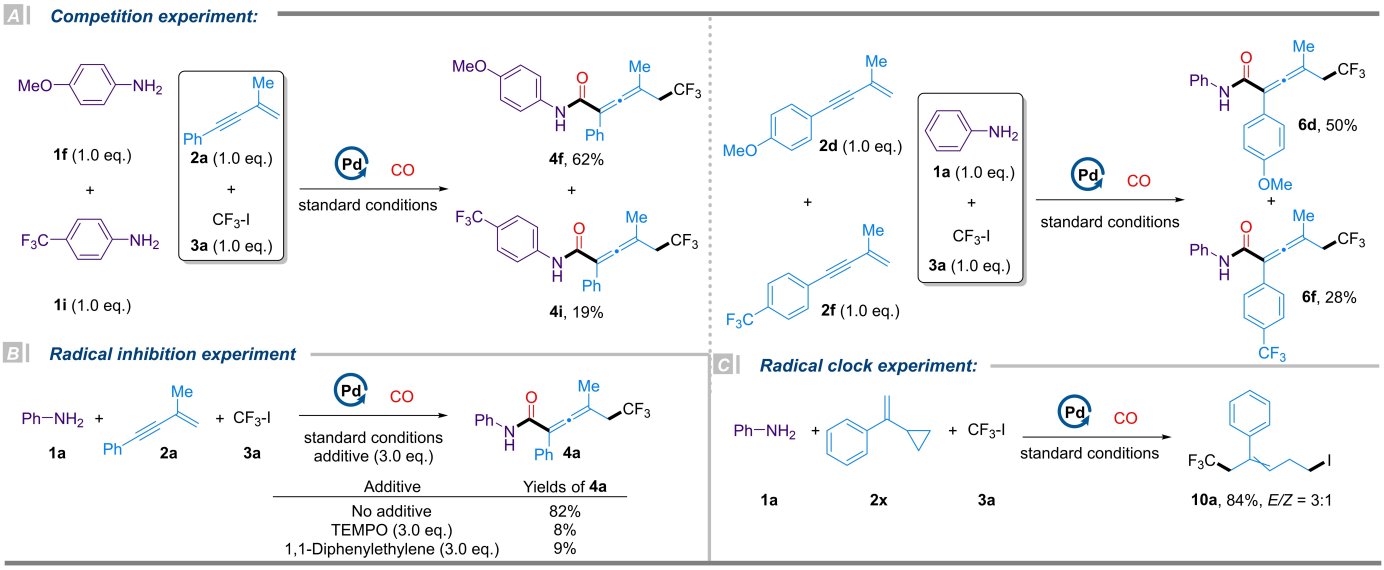
Scheme 5. Synthetic utilizations.

iodo-pyrrol-2-one **9a** and **9b** with high efficiency. In the presence of $FeCl_3$, paraformaldehyde and TfOH, 4-iodo-furan-2(5*H*)-one **9c** could be obtained in 56 % yield. Furthermore, allene ester **5a** could be converted to 4-bromo-2(5*H*)-furanone **9d** in the in the presence of $CuBr_2$ in 62 % yield through bromolactonisation.[11] With the assistance of $FeCl_3$, furan-2(5*H*)-one **9e** could be obtained through cyclization reaction. Treatment with Et_2Zn afforded cyclobutenone (**9f**) with 55 % yield via a stepwise Michael addition/cyclization reaction.[12] The iodo-pyrrol-2-one **9b** can be efficiently constructed using a one-pot two-step method in 80 % yield, further demonstrating the applicability of this approach (Scheme 5D).

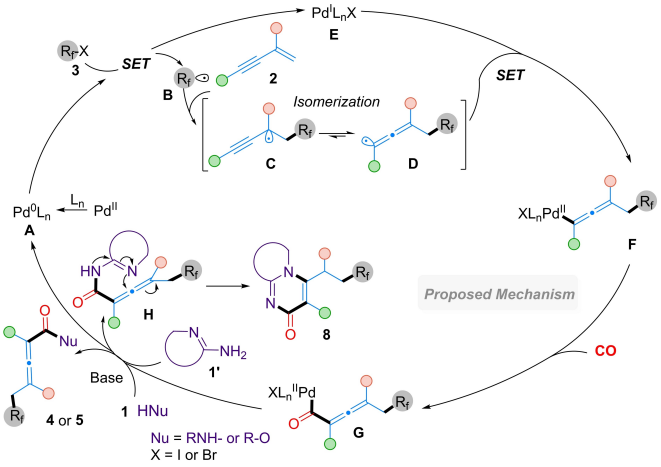
To gain some insight into the mechanism of the multi-component fluoroalkylative carbonylation of 1,3-enynes, a series of controlled experiments were conducted, as depicted in Scheme 6. First, different competitive experiments were carried out separately (Scheme 6A). An evaluation of the reactivities of anilines **1f** and **1i** resulted in the corresponding products **4f** and **4i** with yields of 62 % and 19 %, respectively. This phenomenon may be attributed to the fact that electron-rich nucleophile (**1f**) exhibit faster nucleophilic capabilities compared to electron-deficient substrate (**1i**). When different 1,3-enynes reacted simultaneously with aniline (**1a**) and iodotrifluoromethane (**3a**), more electron-rich product

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Scheme 6. Mechanistic studies.



Scheme 7. Proposed mechanism.

6d was more inclined to be formed over **6f**. This perhaps resulted from the stabilization of carbon radicals by increasing electron density. Next, free radical inhibition experiments were performed. Either TEMPO or 1,1-diphenylethylene was added into the reaction mixtures as a radical scavenger under the standard conditions, respectively, the yield of product **4a** decreased sharply (Scheme 6B). Moreover, radical clock experiment conducted with cyclopropane **2x** resulted in only the corresponding ring-opening fluoroalkyl iodide product **10a** (Scheme 6C). Those results indicate that radical intermediates were most likely involved.

Based on our mechanistic studies and related literatures,[9e,i,13] a palladium-mediated radical mechanism is proposed (Scheme 7). The catalytic cycle initiates with the formation of the active catalyst Pd^0Ln species **A**, generated in situ from the $Pd(COD)Cl_2$. Then, complex **A** induced a single-electron transfer (SET) reduction of Rf-X, leading to the generation of the fluoroalkyl radical **B** and Pd^ILnX species **E**. Subsequently, the fluoroalkyl radical **B** was captured by enyne **2** to yield a tertiary propargyl radical **C**, which can be isomerized to relatively stable allenyl radical **D**. Following this, the Pd^ILnX species **E** reincorporated with the allenyl radical **D** to afford the Pd^{II} complex **F**, which undergoes insertion of carbon monoxide to provide intermediate **G**. Finally, intermediate **G** reacts with nucleophile **1** to produce the desired fluoroalkylative carbonylation products (**4** or **5**) and regenerate Pd^0Ln for the next catalytic cycle with the assistance of base. Notably, when employed bisnucleophile **1′** reacts with intermediate **G** to afford intermediate **H**, which underwent nucleophilic addition with allene to give the cyclization products **8**.

In conclusion, we have developed an efficient palladium-catalyzed multi-component fluoroalkylative carbonylation of 1,3-enynes. A broad range of fluoroalkyl substituted allene carboxylic derivatives were prepared successfully with high chemo- and regio- selectivities and outstanding functional group tolerance. Additionally, more intricate fluoroalkyl-substituted pyrimidinones can be constructed by simply altering the 1,3-bisnucleophilic reagent. This approach also provides a valuable strategy for the late-stage modification of naturally occurring molecules and concisely constructing diverse cyclic compounds.

Acknowledgements

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: 1,3-Enyne · Carbonylation · Domino Reaction · Multicomponent Reaction · Palladium

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